"On my honor. I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues"

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(1)

$$x_1 + \alpha x_2 - y x_3 = \alpha$$

$$3\alpha x_1 + 5 x_2 - 9y x_3 = 1$$

$$2x_1 + \beta x_2 + 3 x_3 = \alpha^2$$

$$x_1 + 4x_2 - 6 x_3 = 2$$

2)

a)

b)

c) $x + 4y - z = 6$ $\quad y, z, a, b$ are free variables

$x = 6 - 4y + z$

$K = \begin{bmatrix} -4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$A = \begin{bmatrix} -4 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

column space of matrix A is K

4 dimensional space

3) a)

Suppose, u is a unit vector in R^n. There, H is the $n \times n$ symmetric matrix such that, $H = I - 2uu^T$

u is unit vector so $\|u\| = 1$

Now, $H^2 = H \cdot H = (I - 2uu^T)(I - 2uu^T)$

$= I^2 - I \cdot 2uu^T - I \cdot 2uu^T - 4uu^T uu^T$

$= I - 2uu^T - 2uu^T + 4u \|u\|^2 u^T$

$\|u\|^2 = u^T u$

$\|u\| = \sqrt{u^T u}$

$\|u\| = 1$

$H^2 = I - 4uu^T + 4uu^T$

$\therefore H^2 = I$

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b) $HH^T = (I - 2uu^T) \cdot (I - 2uu^T)^T$

$= (I - 2uu^T)(I^T - (2uu^T)^T)$ $\quad [(A-B)^T = A^T - B^T]$

$= (I - 2uu^T)(I - 2(u^T)^T u^T)$ $\quad [(AB)^T = B^T A^T]$

$= (I - 2uu^T)(I - 2uu^T)$ $\quad [AB(u^T)^T = u]$

$= I^2 - 2uu^T - 2uu^T + 4uu^T uu^T$

$= I - 4uu^T + 4u \|u\|^2 u^T \quad \|u\| = 1$

$= I - 4uu^T + 4uu^T$

$= I$

similarly, $H^T H = I$

$HH^T = H^T H = I$

so, H is an orthogonal matrix

3) c) For matrix A, a nonzero vector will be eigen vector of A corresponding eigen value λ $\quad Av = \lambda v$

$$Hv = (I - 2vv^T)\cdot v$$
$$= Iv - 2vv^Tv$$
$$= v - 2v\cdot \|v\|^2 \qquad \text{AB I is identity}$$

So, 1 is the only eigen value of I and, $Iv = 1\cdot v = v$ and $\|v\| = +\sqrt{v^Tv}$ v is a unit vector. $\|v\| = 1 = +\sqrt{v^Tv}$

$$v^Tv = 1 = \|v\|^2$$

$$Hv = v - 2v\cdot 1 = -v$$

$$Hv = -v = (-1)\cdot v$$

we have $\Rightarrow Hv = \lambda v \qquad \lambda = -1$

so, v is an eigen vector of H and corresponding eigen value for v is equal to $\lambda = -1$

d) Given, $H = (I - 2vv^T)$

$$H^T = (I - 2vv^T)^T$$
$$= I^2 - 2(vv^T)^T \quad \left[(A-B)^T = A^T - B^T\right]$$
$$= I - 2(v^T)^Tv^T \quad \left[(AB)^T = B^T\cdot A^T\right]$$
$$= I - 2vv^T = H$$

so, H is a symmetric matrix and there is a theorem which states that,

$H = Q\Lambda Q^T$ (Q is an orthogonal matrix)

$= Q\Lambda Q^{-1}$ $\quad (QQ^T = I \Rightarrow Q^T = Q^{-1})$

$= (Q^{-1})^{-1}\Lambda Q^{-1}$ $\quad \Lambda =$ (Diagonal matrix consists of eigen values of H)

$= P^{-1}\Lambda P$

so, H is diagonalizable matrix